DSA SCHEDULE

Beginner Pacing Curriculum Aligned to CSE2002 College Syllabus (CAT-1: Mod 1-2 | CAT-2: Mod 3-4 | TEE: All Mod)

Nov 05, 2026 – May 12, 2027

189 Days	18 Modules	119 Days	33 Days	37 Days
TOTAL SPAN	FULL CURRICULUM	CORE PREP	DEDICATED REVISION	EXAM WINDOWS (3X)

MODULE & TOPIC FOCUS	DIFFICULTY	CORE PREP	REVISION	TOTAL	CALENDAR WINDOW	COLLEGE CSE2002 SYLLABUS ALIGNMENT
21. OOPs in Java College Unit 1 Foundation	Medium	5 Days	2 Days	7 Days	Nov 05 – Nov 11, 2026	Classes, objects, inheritance, polymorphism, encapsulation (ADT basis).
25. Time & Space Complexity College Unit 1 (Complexity)	Easy-Med	3 Days	1 Days	4 Days	Nov 12 – Nov 15, 2026	Big-O, Omega, Theta asymptotic bounds, recurrence analysis, loop parsing.
22-23. Recursion (Part 1 & 2) Foundation for D&C	Hard	8 Days	3 Days	11 Days	Nov 16 – Nov 26, 2026	Stack frames activation, recursion tree drawing, recurrence relations.
24. Divide & Conquer + Sorting College Unit 2 (Sorting & Searching)	Medium	6 Days	2 Days	8 Days	Nov 27 – Dec 04, 2026	Bubble, Selection, Insertion, Merge, Quick Sort & Linear/Binary Search.
27. ArrayLists College Unit 1 (ADT Classification)	Easy	3 Days	1 Days	4 Days	Dec 05 – Dec 08, 2026	Dynamic sizing mechanics, multi-dimensional lists, Linear ADT modeling.
26. Backtracking Advanced Algorithmic Paradigm	Hard	6 Days	2 Days	8 Days	Dec 09 – Dec 16, 2026	State-space trees, branch pruning, N-Queens, Sudoku solver logic.
🛑 **CAT-1 Examination Window (LOCKED)**				8 Days	Dec 17 – Dec 24, 2026	**0 DSA Days — 100% University Exam Focus & GPA Shield.**
28-29. Linked Lists (Part 1 & 2) College Unit 3 (List ADT)	Med-Hard	7 Days	2 Days	9 Days	Dec 25 – Jan 02, 2027	Singly, Doubly, Circular Linked Lists, node reversals, Floyd's cycle, Array vs LL.
30-31. Stacks (Part 1 & 2) College Unit 3 (Stack ADT)	Med-Hard	5 Days	2 Days	7 Days	Jan 03 – Jan 09, 2027	Array & LL implementations, Infix to Postfix, Monotonic stack, applications.
32. Queues College Unit 3 (Queue ADT)	Medium	4 Days	1 Days	5 Days	Jan 10 – Jan 14, 2027	FIFO dynamic buffers, Circular Queues, Deques, Queue using Stacks.
34-36. Binary Trees (Part 1, 2 & 3) College Unit 4 (Trees)	Hard	8 Days	2 Days	10 Days	Jan 15 – Jan 24, 2027	Tree terminologies, Expression Trees, In/Pre/Post/Level traversals, views.
37-38. Binary Search Trees (Part 1 & 2) College Unit 4 (BST & Balanced Trees)	Medium	4 Days	2 Days	6 Days	Jan 25 – Jan 30, 2027	BST search/insert/delete, validation traits, AVL/Red-Black Tree concepts.
40. Hashing College Unit 4 (Hashing)	Medium	4 Days	1 Days	5 Days	Jan 31 – Feb 04, 2027	Hash functions, Separate Chaining, Open Addressing collision resolution.
🛑 **CAT-2 Examination Window (LOCKED)**				8 Days	Feb 05 – Feb 12, 2027	**0 DSA Days — 100% University Exam Focus & GPA Shield.**
39. Heaps / Priority Queues Prerequisite for Unit 5 Graph Algos	Med-Hard	5 Days	2 Days	7 Days	Feb 13 – Feb 19, 2027	Binary Min/Max heaps, heapify algorithm, priority queue hooks for Dijkstra/Prim.
42-47. Graphs (Part 1-5 + Supplemental) College Unit 5 (Graph ADT & MST)	Very Hard	14 Days	4 Days	18 Days	Feb 20 – Mar 09, 2027	Adjacency Matrix/List, BFS/DFS, Topological Sort, Dijkstra, Floyd, Prim, Kruskal.
📚 **College DSA Semester TEE Consolidation** TEE All 5 Units Review	Medium	1 Days	—	1 Days	Mar 10 – Mar 10, 2027	Comprehensive university mock tests, proof derivations & theory revision.
🛑 **TEE Examination Window (LOCKED)**				21 Days	Mar 11 – Mar 31, 2027	**0 DSA Days — 100% University Exam Focus & GPA Shield.**
33. Greedy Algorithms Core Industry LeetCode Pattern	Medium	5 Days	2 Days	7 Days	Apr 01 – Apr 07, 2027	Fractional Knapsack, job sequencing, activity selection, interval heuristics.
41. Tries Advanced Retrieval Structure	Medium	5 Days	2 Days	7 Days	Apr 08 – Apr 14, 2027	Prefix tree construction, startsWith queries, autocomplete search models.
48-53. Dynamic Programming (Part 1-6) Tier-1 Tech Interview Dominance	Very Hard	17 Days	4 Days	21 Days	Apr 15 – May 05, 2027	Memoization to Tabulation, 0-1 Knapsack, LCS, LIS, Catalan Numbers, MCM.
54. Segment Trees Advanced Competitive Programming	Hard	5 Days	2 Days	7 Days	May 06 – May 12, 2027	Range sum/min queries, point updates, lazy propagation mechanisms.

Academic Policy Note: CAT-1 (8 Days), CAT-2 (8 Days), and TEE (21 Days) have zero DSA tasks allocated to protect your GPA. The schedule sequence is restructured so that Units 1 & 2 conclude prior to CAT-1, Units 3 & 4 conclude prior to CAT-2, and Unit 5 concludes prior to TEE.